Microeconomic Analysis: Problem Set I

Due to week 2

1 Problem: Dot Product – Law of Demand

Consider a consumer choice problem with n goods. A demand correspondence $\mathbf{x}(\mathbf{p})$, $\mathbf{x}, \mathbf{p} \in \mathbb{R}^n_+$ is said to satisfy the Law of Demand if for any $\mathbf{p}, \mathbf{p}' \in \mathbb{R}^n_+$, $\mathbf{p} \neq \mathbf{p}'$ the following inequality holds:

$$[\mathbf{x}(\mathbf{p}) - \mathbf{x}(\mathbf{p}')] \cdot [\mathbf{p} - \mathbf{p}'] < 0$$

For Cobb-Douglas preferences we get $x_i = \alpha_i m/p_i$ with $\sum_{i=1}^n \alpha_i = 1$. Check whether the Law of Demand holds for Cobb-Douglas demands.

2 Problem: Linear System. Submit this problem for marking.

Consider the system of linear equations:

$$x + 3y - 2z = 2$$

$$x + 2y + z = 1$$

$$x + 5y + \alpha z = \beta.$$

1. Compute, for each possible value of the parameters α and β, the solutions of the system.
2. Using your answer to the previous question, can you say whether the vector $\begin{pmatrix} 2 \\ 1 \\ \beta \end{pmatrix}$ is spanned or not by vectors $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 1 \\ \alpha \end{pmatrix}$?

3 Problem: Span

Consider the vectors $\mathbf{u} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$.

1. Compute the set $W = \{\mathbf{w} \in \mathbb{R}^3 : span[\mathbf{u}, \mathbf{v}] = span[\mathbf{u}, \mathbf{v}, \mathbf{w}]\}$.

2. Can you explain why W coincides or not with $span[\mathbf{u}, \mathbf{v}]$?
3. Let $\mathbf{z} \notin W$. Show that $\mathbf{u}, \mathbf{v}$ and $\mathbf{z}$ form a basis of $\mathbb{R}^3$.

4 Problem: Some Results from the Lecture

1. Let A be $n \times m$ matrix and B be $m \times l$ matrix. Show that $(A \cdot B)^T = B^T \cdot A^T$.
2. Let
$$A = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ 0 & 2 & 3 & \dots & n \\ 0 & 0 & 3 & \dots & n \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & n \end{pmatrix}$$
Compute $\det A$.

5 Problem: Rank and Span

Consider the vectors
$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{u}_2 = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}, u_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, u_4 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, u_5 = \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}$$

1. Compute the rank of the 3×5 matrix formed by these vectors.
2. Use your answer to the previous question to determine whether these vectors span $\mathbb{R}^3$ or not. Can you describe the vectors that are not spanned by $u_1, \dots, u_5$?
3. Prove, without computations, that all the square matrices composed by three of the five vectors above have the same determinant.

6 Eigenvectors and eigenvalues

Consider the matrix $A = \begin{pmatrix} -1 & 2 \\ -6 & 6 \end{pmatrix}$.

1. Without computing the exact values of the eigenvalues of A, can you say what is their sign?
2. Use the determinant and trace of A to compute its eigenvalues.
3. Use the characteristic polynomial method to find the eigenvalues and eigenvectors of the matrix and show that you can construct a matrix S such that $S^{-1}AS = D$, where D is the diagonal matrix formed by the eigenvalues of A.
4. Compute A^{10}.